AgriShield-X: PSO-Optimized LightGBM Framework for Real-Time Agricultural Drought Forecasting Using Multi-Source Satellite and Climate Data

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Abstract—Agricultural drought is a slow-onset hazard that affects crop growth, irrigation demand, and food security before its impact becomes visible at field scale. This paper presents AgriShield-X, a real-time agricultural drought forecasting framework for strict next-month prediction of the Vegetation Health Index (VHI). The study focuses on the Sindh region and uses a multi-source satellite and climate dataset extracted through Google Earth Engine for 2001–2023. The final dataset contains 1,361,299 monthly grid-level records across 4,937 grid cells and includes vegetation, thermal, rainfall, atmospheric, seasonal, SPI, SPEI, solar radiation, humidity, wind speed, and available soil moisture features. The proposed model uses Particle Swarm Optimization (PSO) to tune a LightGBM regressor under a strict chronological validation protocol. The target variable is vhi_next_month, and target-month VHI, VCI, and TCI are excluded from the input features to prevent leakage. The final model achieved R^2 = 0.8153, RMSE = 0.1097, MAE = 0.0839, and drought severity F1-score = 0.6354. AgriShield-X is also implemented as a cloud-ready decision-support system using FastAPI, React, Docker, AWS EC2, AWS S3, Terraform, authentication, map visualization, crop recommendation, and farmer advisory modules.

Keywords—agricultural drought forecasting, VHI, LightGBM, Particle Swarm Optimization, Google Earth Engine, remote sensing, SPI, SPEI, cloud deployment

I. INTRODUCTION

Climate variability has increased the frequency and uncertainty of agricultural drought in many semi-arid regions. Unlike short meteorological anomalies, agricultural drought develops through the interaction of rainfall deficit, temperature stress, vegetation degradation, evapotranspiration demand, and soil moisture depletion. Its effects are directly experienced by farmers through reduced germination, crop stress, lower yield, and increased irrigation cost. Because drought develops gradually, early detection and next-month forecasting are more valuable than retrospective monitoring alone.

Satellite remote sensing has become a practical source of drought information because it provides repeated observations over large regions where dense ground stations are unavailable. Vegetation indices such as the Normalized Difference Vegetation Index (NDVI), thermal indicators such as land surface temperature (LST), and composite indicators such as the Vegetation Health Index (VHI) are widely used to describe vegetation stress [1]–[3]. However, many existing drought systems primarily estimate present conditions. A farmer-facing warning platform must answer a more difficult question: what is likely to happen next month?

This paper proposes AgriShield-X, a complete drought forecasting and deployment framework for the Sindh region. The framework predicts next-month VHI using only information available from historical and current months. The central design principle is research honesty. Same-month drought reconstruction and random split experiments can produce high apparent accuracy, but they do not represent operational future forecasting. Therefore, this work reports strict chronological next-month forecasting as the main result.

The proposed system combines multi-source Google Earth Engine data, climate feature engineering, drought index construction, PSO-based hyperparameter optimization, and LightGBM regression. Beyond model training, AgriShield-X includes a deployed FastAPI backend, React frontend, AWS infrastructure, authentication, forecast dashboard, interactive drought map, explainability view, crop recommendation, and farmer advisory page. The project therefore bridges the gap between a research model and a usable agricultural decision-support prototype.

The Sindh region presents a useful case study because it contains irrigated agriculture, arid and semi-arid zones, coastal influence, and strong seasonal dependence. A model that performs reasonably across thousands of grid cells in this setting must learn both spatial and temporal variation. This motivated the use of grid-level records rather than a single aggregated regional time series.

The contributions of this work are: (1) construction of a large Sindh drought forecasting dataset from 2001 to 2023; (2) leakage-safe next-month target generation using vhi_next_month; (3) PSO-based optimization of LightGBM for strict chronological forecasting; (4) evaluation using regression and drought severity metrics; and (5) end-to-end deployment of AgriShield-X on cloud infrastructure.

II. LITERATURE REVIEW

Drought monitoring has traditionally relied on statistical indices such as SPI, SPEI, Palmer-type drought measures, and vegetation-condition products [4], [5]. These indices are interpretable and useful for long-term monitoring, but they are not always sufficient for future forecasting because agricultural drought depends on multiple interacting variables. Rainfall deficit may not immediately cause crop stress if soil moisture remains available, while high temperature and solar radiation may intensify drought even when rainfall is near normal.

Machine learning methods have been applied to drought prediction because they can learn non-linear relations between vegetation, rainfall, temperature, and seasonal behavior [6]. Random Forest, ExtraTrees, Support Vector Regression, XGBoost, and LightGBM have been widely used for environmental modeling and tabular forecasting tasks [7]–[9]. Gradient boosting models are especially suitable for heterogeneous satellite-climate data because they handle non-linear thresholds, feature interactions, and missing values more effectively than linear models.

Deep learning methods such as LSTM, GRU, CNN-LSTM, and attention-based networks have also been studied for drought forecasting [10], [11]. These methods can learn temporal dependencies, but they require dense continuous sequences and careful leakage control. In grid-level drought forecasting, the number of months per grid cell is often modest even when the total row count is large. This makes robust tabular gradient boosting a strong practical baseline.

Hybrid and heuristic models attempt to improve performance by combining feature engineering, ensemble learning, and optimization algorithms. Particle Swarm

Optimization, Genetic Algorithm, Grey Wolf Optimizer, Whale Optimization Algorithm, and Bayesian optimization are often used to tune model parameters or feature subsets [12], [13]. PSO is attractive because it is simple, derivative-free, and suitable for mixed search spaces where model complexity and sampling parameters must be optimized together.

Remote sensing drought studies often differ in their definition of the prediction target. Some estimate vegetation status for the same month using other indicators observed in that month, while others forecast a future index from previous observations. These tasks are not equivalent. Same-month estimation is useful for gap filling and reconstruction, but it is easier because the model observes signals from the target period. Future forecasting is more difficult because rainfall and heat anomalies can change after the input window has ended.

Another issue is spatial generalization. A random split can place records from the same grid cell and adjacent months in both training and testing sets. This makes the test distribution very similar to the training distribution and can inflate reported metrics. Chronological splitting reduces this problem by forcing the model to extrapolate from previous years to later months. AgriShield-X adopts this stricter setting for its primary result.

A key gap in the literature is evaluation protocol. Many studies report strong results using random split validation or same-month reconstruction. Such protocols may unintentionally allow spatial or temporal leakage and can overstate real future forecasting performance [14]. AgriShield-X addresses this gap by using strict chronological next-month prediction as the main claim and by clearly separating it from easier estimation settings.

PLACEHOLDER: System Architecture GEE Data → Feature Engineering → PSO-LightGBM → FastAPI → React Dashboard

Fig. 1. System architecture of AgriShield-X.

III. PROPOSED METHODOLOGY

The proposed methodology consists of data acquisition, preprocessing, feature engineering, target generation, PSO optimization, LightGBM training, and cloud deployment. Each stage was designed to preserve temporal causality. The forecasting model is not allowed to observe target-month VHI, VCI, or TCI when predicting the next month.

Google Earth Engine was used to extract monthly grid-level observations for Sindh. The sources include MODIS NDVI, MODIS LST, CHIRPS rainfall, ERA5 climate variables, EVI, solar radiation, temperature, humidity, wind speed, and available soil moisture. SPI and SPEI features were generated at 1-, 3-, 6-, and 12-month scales to represent rainfall anomaly and climatic water-balance stress.

Preprocessing included date normalization, grid identifier cleaning, duplicate removal, missing value handling, and temporal alignment. Each row corresponds to a grid cell and a month. Missing values caused by source availability were retained where meaningful and imputed where required for model training. This approach avoids discarding entire months when a single auxiliary data source is unavailable.

VHI was generated from vegetation and thermal condition information. The next-month target was then constructed by shifting VHI one month forward for each grid cell. This produces the supervised target vhi_next_month. The feature matrix contains only present and past-month predictors, including lag features, rolling means, rolling standard deviations, seasonal encodings, SPI, SPEI, and climate variables.

PSO was used to optimize LightGBM hyperparameters. Each particle represented a candidate LightGBM configuration. The particle position encoded values such as number of estimators, learning rate, tree depth, number of leaves, subsampling, column sampling, regularization, and minimum child samples. The objective function minimized validation RMSE on a chronological validation split. After PSO convergence, the best configuration was retrained and evaluated on future holdout data.

For each grid cell, temporal features were calculated independently so that rolling values were not mixed across locations. Short-window statistics represent immediate weather and vegetation stress, while longer windows capture seasonal drought memory. The month sine and cosine variables were included because the same raw rainfall value can have different agricultural meaning in pre-monsoon, monsoon, and post-monsoon periods.

The PSO search was preferred over manual tuning because the feature set contains variables with different temporal behavior and missingness patterns. A shallow model may underfit spatial heterogeneity, while an overly complex model may memorize grid-specific noise. The swarm search therefore balances tree complexity, shrinkage, subsampling, and regularization using validation RMSE as the fitness score.

LightGBM was chosen for the final production candidate after comparing several booster families during project development. It trains efficiently on large tabular data, supports non-linear feature interactions, and gives stable performance under imputed climate features. This made it more practical for cloud deployment than heavier sequence models that require larger memory and longer inference time.

A separate validation pass was performed after the Google Earth Engine export stage. Early exports contained empty monthly files, duplicate grid-date rows, and missing climate bands. The corrected pipeline was designed to keep base grid rows even when some optional sources were unavailable, attach availability flags, and compute VHI and vhi_next_month after duplicate removal. This step was necessary because model accuracy is strongly affected by data integrity; a powerful algorithm cannot recover from broken temporal alignment.

The final cleaned dataset was checked for row count, grid count, date range, duplicate grid-date records, missing values, target availability, and climate-variable completeness. This audit also helped identify which variables were most promising for forecasting. Solar radiation, temperature trends, seasonal encodings, and vegetation rolling features were consistently meaningful, while sparsely available features such as soil moisture were useful only when safely imputed and not treated as complete observations.

Leakage prevention was treated as a methodological requirement rather than a post-processing check. The target was constructed only after sorting each grid cell by month. Lag and rolling features were computed from current or previous observations. The strict feature set excluded target-month VHI, target-month VCI, and target-month TCI. This design ensures that the model cannot obtain the answer indirectly from the same month that it is asked to predict.

In addition, the research dashboard reports strict forecasting metrics separately from same-month estimation

metrics. This separation is important for scientific clarity. Same-month estimation is still useful for understanding whether the engineered features can reconstruct drought state, but it cannot be used as the headline future-forecasting claim. AgriShield-X therefore uses strict next-month forecasting as the production candidate even though easier protocols can produce higher values.

The strict forecasting pipeline follows five rules: records are sorted by date, future rows are held out, no random shuffling is used for the main result, target-month VHI-derived variables are excluded from inputs, and performance is reported only on unseen later months. These rules make the experiment closer to real-time deployment.

PLACEHOLDER: Data Pipeline MODIS + CHIRPS + ERA5 + SPI/SPEI → Cleaning → Lag/Rolling Features → vhi_next_month

Fig. 2. Data processing pipeline.

PLACEHOLDER: PSO-LightGBM Workflow Particles → Hyperparameters → Chronological Validation RMSE → Global Best Model

Fig. 3. PSO-LightGBM workflow.

IV. EXPERIMENTAL SETUP

The experiments were performed using Python-based machine learning libraries. Data processing used pandas and NumPy; model training used LightGBM and scikit-learn; optimization used a custom PSO procedure; and trained artifacts were stored using joblib. The backend system was developed using FastAPI and SQLAlchemy, while the frontend was implemented using React and Vite. Docker was used for containerized deployment.

The dataset covers the Sindh region from 2001 to 2023 and contains 1,361,299 grid-month records across 4,937 grid cells. The main split was chronological: earlier months were used for training and validation, while later months were reserved for testing. This design evaluates whether the model can predict future drought behavior rather than memorize current-month conditions.

Before training, candidate leakage columns were removed from the strict forecasting feature matrix. The model was allowed to use lagged and rolling VHI-derived values because these represent historical drought state, but it was not allowed to use the target month values. This distinction is important because drought indices are autocorrelated; using the wrong temporal alignment can create unrealistically high accuracy.

The experiment also produced model artifacts, metric reports, and diagnostic plots. These artifacts were kept separate from the data-expansion pipeline so that retraining and production serving could be controlled independently. This separation reduces deployment risk and helps reproduce the exact forecast result shown in the dashboard.

Four evaluation metrics were used. R^2 measures explained variance, RMSE penalizes larger prediction errors, MAE measures average absolute deviation, and F1-score evaluates drought severity class prediction after converting continuous VHI forecasts into drought classes. RMSE was prioritized during PSO because large drought-index errors can lead to incorrect risk alerts.

TABLE I. DATASET SUMMARY

Attribute	Value
Region	Sindh
Study period	2001–2023
Rows	1,361,299
Grid cells	4,937
Resolution	Monthly grid-cell records
Target	vhi_next_month
Forecasting protocol	Strict chronological next-month

TABLE II. FEATURE DESCRIPTION

Feature group	Purpose
NDVI/EVI	Vegetation vigor and greenness
LST/temperature	Thermal stress and heat anomaly
CHIRPS rainfall	Monthly precipitation signal
SPI/SPEI	Multi-scale rainfall and water-balance anomaly
Solar radiation	Radiative and evapotranspiration stress
Humidity/wind speed	Atmospheric drying conditions
Lag/rolling features	Temporal memory of drought state
Month sine/cosine	Seasonality representation

TABLE III. PSO-OPTIMIZED HYPERPARAMETER GROUPS

Parameter group	Optimized role
n_estimators	Boosting rounds
learning_rate	Step size control
num_leaves/max_depth	Tree complexity
subsample/colsample	Row and feature sampling
reg_alpha/reg_lambda	Regularization
min_child_samples	Leaf stability
PSO objective	Minimize chronological validation RMSE

V. RESULTS AND DISCUSSION

The final strict next-month forecasting result is shown in Table IV. The PSO-Optimized LightGBM model achieved $R^2 = 0.8153$, RMSE = 0.1097, MAE = 0.0839, and drought severity F1-score = 0.6354. These values represent true future forecasting performance, not same-month reconstruction. The result is therefore more conservative than random split or reconstruction protocols, but it is more meaningful for operational drought warning.

The R^2 score indicates that the model explains a substantial fraction of future VHI variation across the study region. The remaining error is expected because agricultural drought is influenced by future rainfall events, irrigation practices, crop type, soil properties, and local management decisions that are not fully captured by satellite variables alone. The RMSE and MAE values show that the model is useful for risk ranking and early warning, while still leaving room for improvement in exact VHI prediction.

Feature importance analysis showed that seasonal encodings, solar radiation, temperature rolling averages, VCI rolling indicators, VHI rolling indicators, latitude, and short-term solar-radiation trends were among the strongest contributors. This pattern is physically reasonable. Seasonal features capture monsoon and crop-cycle timing, radiation and temperature represent atmospheric demand, and rolling vegetation variables capture delayed plant response to previous stress.

For farmers, the practical value of the result is not limited to a single numeric forecast. The system converts predicted VHI into severity categories, risk score, confidence score, crop recommendation, and advisory messages. Early warning can support irrigation scheduling, selection of drought-tolerant crops, and risk-aware farming decisions. For administrators, grid-level forecasts can help prioritize regions for field inspection and water-resource planning.

Compared with earlier project baselines, the enriched GEE dataset improved strict forecasting by adding climate stress variables that were absent from the first dataset. In particular,

solar radiation, humidity, temperature trend, and SPI/SPEI features helped the model distinguish between vegetation stress caused by normal seasonality and stress caused by persistent drought conditions.

The metric values should be interpreted in the context of operational forecasting. A perfect or near-perfect R^2 is not expected when predicting one month ahead over more than two decades and thousands of grid cells. Sudden rainfall events, irrigation releases, cloud contamination, crop changes, and local management choices can alter future VHI after the model input date. Therefore, the goal of AgriShield-X is early risk ranking and advisory support rather than exact physical simulation of every field.

The drought severity F1-score is especially relevant for farmer advisories. A small continuous VHI error may not matter if the severity class remains unchanged, but an error near a class boundary can change the recommended action. For this reason, the frontend reports both numerical forecast indicators and categorical severity labels.

The strict split also changes how model success should be judged. A random split would test whether the model can interpolate among similar grid-month records, while chronological testing asks whether previous climate-vegetation relationships remain useful in future years. This is a harder but more realistic requirement for early warning. The reported score therefore reflects both model quality and temporal stability of the climate signal.

From an engineering perspective, the final result was selected not only by accuracy but also by reliability. The PSO-LightGBM artifact could be loaded consistently in the backend container, exposed through a small prediction service, and served through the frontend without noticeable latency. This mattered because the final-year project required a working prototype rather than an offline notebook result.

The interpretability layer is intentionally simple. Instead of presenting complex mathematical explanations to farmers, the dashboard exposes top factors such as rainfall, temperature, solar radiation, vegetation trend, and seasonal timing. These factors are easier to connect with practical action, such as irrigation planning, crop selection, or risk monitoring. A more advanced version can add SHAP-based local explanations, but the present interface already gives users a readable reason for the forecast.

The model did not claim to outperform very high same-month reconstruction metrics as a future forecast. This distinction is central to the research contribution. AgriShield-X reports the strict future result as the production candidate and treats same-month estimation, where available, as a separate diagnostic task rather than the main forecasting claim.

A deployment-oriented model must also consider inference behavior. The final LightGBM artifact is compact enough to load inside the FastAPI service and respond quickly to dashboard requests. This is important for farmer-facing systems where many users may request forecasts from mobile devices or low-bandwidth networks. A heavier deep sequence model may improve certain patterns, but it would increase memory consumption and complicate deployment on modest cloud instances.

The results further suggest that climate enrichment was more valuable than adding model complexity alone. Earlier experiments using a smaller feature set produced weaker strict forecasting performance. After adding ERA5-derived climate indicators, solar radiation, SPI, SPEI, and humidity-related signals, the model had a better representation of atmospheric stress. This observation supports the view that drought forecasting performance is often limited by data coverage before it is limited by algorithm choice.

Because VHI is partly vegetation-based, the model naturally performs better during periods when vegetation response follows seasonal climate patterns. Errors become larger during abrupt transitions, for example after unusual rainfall, irrigation changes, or localized land-cover disturbance. These cases are important for future field validation because they reveal where satellite-only forecasting should be combined with agronomic ground information.

The numerical metrics were therefore interpreted together with system behavior. A forecast with moderate continuous error may still be useful if it correctly identifies increasing drought risk and provides timely advisory guidance. In contrast, a highly accurate same-month reconstruction would be less useful for planning if it cannot warn farmers before stress occurs. AgriShield-X prioritizes the first use case.

The main limitation is that strict forecasting is inherently harder than same-month estimation. If current-month VHI or target-month VCI/TCI are used as inputs, the model can appear much more accurate, but that setting does not represent future prediction. AgriShield-X intentionally avoids this leakage and reports the more difficult protocol as the main result.

TABLE IV. FINAL PERFORMANCE METRICS

Metric	Value
R^2	0.8153
RMSE	0.1097
MAE	0.0839
Drought severity F1	0.6354
Main protocol	Strict next-month forecasting
PLACEHOLDER: Actual vs Predicted Future holdout VHI compared with PSO-LightGBM prediction	

Fig. 4. Actual versus predicted drought index.

VI. AGRISHIELD-X CLOUD DEPLOYMENT

AgriShield-X was implemented as a complete cloud-ready application. The backend is built with FastAPI and exposes APIs for health checks, benchmark reports, PSO future metrics, authentication, forecasting, drought map data, crop recommendation, farmer advisories, and admin analytics. The model artifacts and reports are loaded by the backend at runtime so the dashboard can display the final research result directly.

The frontend is a React application with an agriculture-focused interface. It includes a landing page, forecast dashboard, interactive Sindh map, historical analysis page, explainability page, crop recommendation page, farmer advisory page, authentication page, admin console, research results page, and Sindh PSO model page. The dashboard separates strict future forecasting from same-month estimation to avoid misleading claims.

The deployment uses AWS EC2 for the backend API and AWS S3 for static frontend hosting. Docker is used to run the backend service, while Terraform files support reproducible infrastructure creation. The security group allows HTTP access and backend API access, while SSH is restricted to the developer IP. An IAM instance profile and SSM support provide safer administrative access. JWT authentication supports ADMIN and FARMER roles.

The backend contains dedicated endpoints for health monitoring, benchmark display, PSO future metrics, authentication, forecast generation, map display, crop recommendation, advisories, and admin analytics. The frontend consumes these endpoints through a single API service layer, which makes it possible to change the deployment host without changing page logic.

Role-based access was implemented to separate farmer functions from administrator functions. Farmers can view forecasts, maps, recommendations, and advisories. Administrators can inspect user counts, active users, forecast usage, model metrics, and system logs. This design is appropriate for a final-year prototype and can be extended to production identity providers in future work.

The deployed system was verified through API smoke tests. The health endpoint confirmed model loading, the PSO future metrics endpoint returned the promoted LightGBM result, authentication endpoints issued JWT tokens, and protected admin analytics were accessible only with an admin token. Forecast, map, crop recommendation, and advisory endpoints were also checked after deployment. These checks are important because a machine learning system is useful only when its trained artifact, API contract, and user interface operate together.

The frontend was built with the deployed backend URL and uploaded to S3 as a static website. Route fallback objects were added for dashboard pages so direct navigation to research results, forecast dashboard, map, and authentication pages remains functional. This makes the prototype easier to demonstrate during evaluation because each page can be opened directly from a browser link.

Although the current deployment is suitable for a final-year prototype, production use would require stronger operational controls. These include managed secrets, HTTPS certificates, automated backups, monitoring alarms, stronger email verification, rate limiting, and a managed database. The current architecture was intentionally kept simple so the research workflow, model serving, and farmer interface remain understandable and reproducible.

This deployment design demonstrates that the project is not only a model-training exercise but also an operational prototype. Farmers can submit drought-related inputs, view severity estimates, inspect map-level risk, obtain crop guidance, and read advisory suggestions through a web interface.

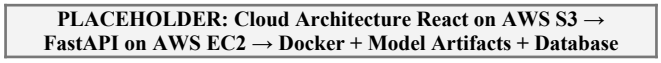


Fig. 5. Cloud deployment architecture.

VII. CONCLUSION AND FUTURE WORK

Several limitations remain. First, the model works at grid-cell resolution and does not know the crop type or irrigation schedule of each field. A district with canal irrigation can recover from rainfall deficit differently from a rainfed area. Second, the satellite variables are monthly aggregates, so short heat waves or short rainfall bursts may be smoothed. Third, some climate variables are derived from reanalysis products rather than direct ground stations, which can introduce uncertainty at local scale.

Another limitation is that VHI is an index-based target rather than a direct yield measurement. VHI is suitable for vegetation health forecasting, but it does not fully represent crop loss, farmer income, or irrigation cost. For decision support, future versions should connect VHI forecasts with crop calendars, ground survey observations, crop yield records, and water-resource information. This would allow the system to move from drought-index prediction toward impact prediction.

The current system nevertheless provides a strong foundation because it integrates the full workflow: data extraction, cleaning, feature engineering, leakage-safe model training, performance reporting, API deployment, authentication, dashboard visualization, and farmer-facing recommendations. Many academic prototypes stop at model metrics; AgriShield-X demonstrates how the model can be converted into a working software system.

For real-world use, deployment monitoring would also be required. Model drift should be checked when new seasons arrive, especially after extreme events. The system should record prediction requests, compare forecasts with later observed VHI, and trigger retraining when error increases. These monitoring functions can be added to the existing admin analytics page and backend reporting pipeline.

This paper presented AgriShield-X, a PSO-Optimized LightGBM framework for strict next-month agricultural drought forecasting using multi-source satellite and climate data. The system was evaluated on a Sindh dataset containing 1,361,299 records and 4,937 grid cells from 2001 to 2023. The final model achieved $R^2 = 0.8153$, RMSE = 0.1097, MAE = 0.0839, and F1-score = 0.6354 under a strict chronological forecasting protocol.

The study shows that enriched satellite and climate variables can improve drought forecasting when combined with careful feature engineering and heuristic hyperparameter optimization. More importantly, it demonstrates the value of honest evaluation. The reported result is lower than what may be obtained through same-month estimation, but it is a more valid estimate of operational next-month forecasting ability.

A further extension is to combine tabular climate features with image patches from Sentinel-2 or other high-resolution sensors. Image patches may capture canal networks, cropped area, water bodies, and localized vegetation stress that are lost in monthly tabular summaries. However, such an extension should still preserve chronological evaluation and avoid target-period leakage.

The current prototype also provides a foundation for field validation. Farmer feedback, district-level crop calendars, irrigation schedules, and yield records can be incorporated to evaluate whether drought risk predictions translate into actionable farm-level outcomes.

The project also highlights a broader lesson for agricultural AI systems: evaluation protocol must match the claim being made. If the claim is forecasting, future data must remain unseen. If the claim is reconstruction, that should be labelled separately. This distinction improves the credibility of the system and helps farmers, examiners, and researchers understand what the model can and cannot promise.

Future work will expand the framework to additional agricultural regions, integrate higher-resolution Sentinel-2 image patches, include ground-level soil and crop information, evaluate transformer-based time-series models, and improve drought-class classification. Mobile access, local-language advisory generation, and integration with irrigation planning systems can further improve farmer adoption.

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